

HOW TO ANALYSE A MUTUAL FUND

Where to Find Fund Information

1. Investment Apps & Brokers

Best for a quick look at risk metrics.

- **Examples:** Groww, Zerodha Coin, IndMoney, ET Money

2. Independent Research Portals

Best for comparing different funds side-by-side.

- **Examples:** Value Research, Morningstar, Moneycontrol

3. Official Documents

Best for the most accurate and final data.

- **Examples:** Monthly Fund Factsheets, AMC Official Websites

Returns Analysis

CAGR — Compound Annual Growth Rate

The standard and most reliable return metric for mutual funds. CAGR tells you how much your money grew each year on average, smoothed out over the entire investment period. It removes the noise of year-to-year market fluctuations and gives you a single clean number to compare

Example: ₹1,00,000 grows to ₹2,00,000 in 6 years → CAGR = 12.25% per year

This means every year, your money compounded at 12.25%. You can now fairly compare this with another fund, a fixed deposit, or a benchmark index.

Check Historical Returns (1-Year, 3-Year, 5-Year, Since Inception)

- When analysing a mutual fund, don't just look at one-year return. Instead, check how it performed over different time periods:
- 1-year return → Shows the short-term performance. Good for knowing how the fund did recently.
- 3-year return → Tells you how the fund handled medium-term ups and downs.
- 5-year return → Gives a clearer picture of long-term consistency.

Check Historical Returns (1-Year, 3-Year, 5-Year, Since Inception)

- Since inception → Shows the fund's overall journey from the day it started.
- **Why this matters?**
- A fund may do very well in one year (due to market boom), but may not perform consistently over 3 or 5 years. Looking at different time frames helps you judge whether the fund is reliable or just lucky.

What CAGR to expect from different funds.

Fund Type	Expected CAGR (Long Term)
Large Cap Fund	10% – 14%
Mid Cap Fund	12% – 18%
Small Cap Fund	15% – 22%
Hybrid Fund	8% – 12%
Debt Fund	6% – 7.5%
Fixed Deposit	6% – 7.5%

Understanding SIP Returns with the SIP Calculator

What is a SIP Calculator? A SIP Calculator is a simple tool that shows you how much your monthly investment will grow over time, based on an expected CAGR. It helps you plan your goals — "How much do I need to invest monthly to reach ₹1 Crore in 15 years?"

Every investment app — Groww, Kuvera, Coin, AMFI website — has a free SIP calculator.

How to Compare Returns Correctly

compare your CAGR with a **Benchmark**.

What is a Benchmark?

A benchmark is a standard yardstick or point of reference used to judge how well your investment is performing. (usually a market index like the Nifty 50) used to measure how well your mutual fund is performing.

compare your CAGR with a **Benchmark**.

If your fund is...	Compare its CAGR with...
Large-Cap Fund	Nifty 50 or BSE Sensex
Mid-Cap Fund	Nifty Midcap 150
Small-Cap Fund	Nifty Smallcap 250
Flexi-Cap / Multi-Cap	Nifty 500
Banking Sector Fund	Nifty Bank
Debt Fund	CRISIL Composite Bond Fund Index

Risk Metrics

Risk metrics are tools used to measure the **"quality"** of a return. They help you understand if a fund's profit came from a solid strategy or just from taking dangerous gambles.

The 5 Key Risk Metrics

1. **Standard Deviation**
2. **Beta**
3. **Sharpe Ratio**
4. **Sortino Ratio**
5. **Alpha**

1. Standard Deviation

This measures how much a fund's performance **swings** away from its average return. It is the primary tool used to measure **volatility**.

- **How it works:** If two funds both give a 15% return, but Fund A has a low Standard Deviation and Fund B has a high one, Fund A is the "smoother" investment. Fund B likely had massive jumps and scary drops to reach that same 15%.
- **Low Standard Deviation:** Means the fund is consistent and steady.
- **High Standard Deviation:** Means the fund is a "rollercoaster"—high highs and low lows.

*It tells you how **bumpy** your investment journey will be. If you prefer a peaceful sleep, you want a fund with a **lower** Standard Deviation.*

2. Beta

This measures how much a fund reacts to the **movement of the market** (the benchmark). It tells you how sensitive your investment is to the overall ups and downs of the stock market.

How to Read Beta Values:

Beta = 1: The fund moves exactly with the market. If the Nifty 50 goes up **10%**, the fund should also go up roughly **10%**.

Beta > 1 (Aggressive): The fund is more volatile than the market. For example, a Beta of **1.2** means the fund is **20%** more volatile than the index. If the market rises, the fund may rise more, but it will also fall harder during a crash.

Beta < 1 (Defensive): The fund is less volatile than the market. A Beta of **0.8** means it is **20%** less volatile. It may rise less during a boom, but it provides better protection when the market falls

3. Sharpe Ratio

The Sharpe Ratio measures the **risk-adjusted return** of an investment. It quantifies how much excess return an investor receives for the extra volatility endured by holding a risky asset rather than a risk-free asset (like a Government Bond).

Sharpe Ratio = (Fund Return - Risk-Free Rate) / Standard Deviation

In short

Sharpe Ratio = Extra Profit / Extra Risk

- **Example:**

Imagine two different funds, and let's assume the **Risk-Free Rate** (like a Government Bond) is **6%**.

Fund A (Conservative)

Actual Return: 12%

**Risk (Standard
Deviation):** 6%

Sharpe ratio :

$$(12\% - 6\%) / 6\% = 1.0$$

Fund B (Aggressive)

Actual Return: 12%

**Risk (Standard
Deviation):** 12%

Sharpe ratio :

$$(12\% - 6\%) / 12\% = 0.5$$

The Conclusion for Investors:

Even though both funds gave the same **12% return**, **Fund A is the better choice**. It has a higher Sharpe Ratio (1.0 vs 0.5), meaning it achieved its profit much more safely. Fund B took twice as much risk for the exact same result.

Key Rule:

Always look for a **higher Sharpe Ratio**. It indicates the fund manager is providing better returns for every unit of risk taken.

4. Sortino Ratio

The Sortino Ratio is a variation of the Sharpe Ratio that differentiates "harmful" volatility from "good" volatility. It only measures the **downside risk** (negative returns) rather than total volatility.

Sortino Ratio = (Fund Return - Risk-Free Rate) / Downside Deviation

In short Sortino Ratio = Extra Profit / **Downside Risk (Losses)**

Example:

Imagine two different funds. Both give a **12% return**, and the Risk-Free Rate is **6%**.

Fund A (Upward Volatility)

- **Performance:** Most months are steady, but the fund had a massive jump of **+20%** in one month. It had **no** negative months.
- **Downside Deviation:** 2% (Very low because the price swings were mostly upward).
- **Sortino Ratio:** $(12\% - 6\%) / 2\% = 3$

Fund B (Downside Volatility)

- **Performance:** The fund is inconsistent. It has months of **+10%** followed by months of **-10%**.
- **Downside Deviation:** 6% (High because the fund frequently drops into negative territory).
- **Sortino Ratio:** $(12\% - 6\%) / 6\% = 1$

The Conclusion for Investors:

Even though both funds gave the same 12% return, **Fund A is the better choice.**

- The **Sharpe Ratio** would penalize Fund A for its +20% jump, calling it "risky."
- The **Sortino Ratio** ignores that upward jump and shows that Fund A is excellent at protecting capital, while Fund B forces you to suffer through many losses.

Key Rule:

Always look for a **higher Sortino Ratio**. It indicates the fund manager is providing returns while successfully avoiding "drawdowns" or significant losses for the investor.

5. Alpha

Definition: Alpha represents the **excess return** an investment generates over its benchmark index. It is the numerical value of the value added (or subtracted) by the fund manager's active strategy.

- **The Calculation:** If a benchmark (like the Nifty 50) returns 12% and the mutual fund returns 15%, the **Alpha is 3%**. Conversely, if the fund returns only 10% in the same period, the Alpha is **-2%**.
- **The Objective:** To measure the specific contribution of the fund manager's stock selection and timing, independent of general market movements.

How to Read Alpha Values:

Positive Alpha (e.g., +2.0): The fund manager has outperformed the benchmark by 2%. This is a sign of a very skilled manager who is "adding value".

Zero Alpha: The fund has performed exactly like the market index. The manager hasn't added any extra value.

Negative Alpha (e.g., -1.0): The fund has underperformed the benchmark by 1%. Even if the fund made a profit, it made less than it should have compared to the market.

Example

Market (Nifty 50) Return: 10%

- **Fund A (Alpha of +3):** Returns **13%**. The manager successfully beat the market by 3%.
- **Fund B (Alpha of -2):** Returns **8%**. Even though the fund is in profit, it performed 2% worse than the market index.

The Verdict: When your students are analyzing a mutual fund, they should look for a **Positive Alpha**. It proves that they are paying the fund manager's "Expense Ratio" for a good reason—because the manager is actually beating the market.

Risk Metrics Summary

Metric	What it measures	Simple Interpretation
Standard Deviation	Total Volatility	Lower is better. It shows how much the price swings. A low number means a smooth and steady ride.
Beta	Market Sensitivity	1.0 is the Market. Above 1.0 means Aggressive (moves more than Nifty). Below 1.0 means Defensive (moves less).
Sharpe Ratio	Overall Efficiency	Higher is better. It shows if the profit is worth the risk. A high number means the manager is smart, not lucky.
Sortino Ratio	Downside Risk	Higher is better. It only looks at bad volatility (losses) . High numbers mean your money is safer during a crash.
Alpha	Manager Skill	Positive is better. It shows the "Extra Profit" the manager made compared to the Nifty index.

THANK YOU